

Carbon Pricing in Reverse: Evidence from Virginia’s Exit and Re-Entry into RGGI

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Abstract

[PENDING DATA: Abstract.] Virginia is the only U.S. state to enter, exit, and re-enter a carbon cap-and-trade program (the Regional Greenhouse Gas Initiative, RGGI) within a five-year span. Because the exit was driven by a change in gubernatorial administration rather than by economic conditions, the timing of treatment is plausibly exogenous. We use synthetic control and event-study difference-in-differences methods on public data to estimate the causal effect of Virginia’s RGGI participation on retail electricity prices, wholesale prices, and CO₂ emissions, and we test whether the effects of *exiting* a carbon market are symmetric to the effects of *entering* it. [PENDING DATA: Headline estimates for the entry (Jan. 2021) and exit (Dec. 2023) events; the re-entry event (Jul. 2026) is analyzed prospectively.]

Keywords: carbon pricing; cap-and-trade; RGGI; synthetic control; electricity markets; emissions leakage; policy reversal.

JEL Codes: Q48, Q54, Q58, L94, H23.

Contents

1	Introduction	3
2	Institutional Background	4
2.1	RGGI Overview	4
2.2	Virginia’s RGGI Timeline	5
2.3	The Dominion Energy Context	5
3	Literature Review	6
4	Theoretical Framework	6
4.1	A Carbon Adder in a Regulated Market	7
4.2	Predictions	7
5	Data	7
5.1	Sample Construction	8
6	Empirical Strategy	9
6.1	Synthetic Control (Primary)	9
6.2	Event-Study Difference-in-Differences	9
6.3	Synthetic Difference-in-Differences	9
6.4	Extensions	9
7	Threats to Identification and Robustness	10
8	Results	11
8.1	Exit Event (Dec. 2023)	11
8.2	Entry Event (Jan. 2021)	11
8.3	Asymmetry Test	11
8.4	Emissions Leakage	11
9	Expected Results and Interpretation	12
10	Conclusion	12
A	Data Sources and Public Retrieval Methods	16
B	Pre-Analysis Plan: Re-Entry (Jul. 2026)	16

1 Introduction

Cap-and-trade is the dominant market-based instrument for regulating greenhouse-gas emissions in the United States, yet credible causal evidence on its effects remains surprisingly thin. The difficulty is one of inference rather than data: carbon markets are adopted by jurisdictions that differ systematically from non-adopters, and they are rarely turned off, so the analyst seldom observes a clean counterfactual for what a participating economy would have looked like absent the policy. Most of what we know about the Regional Greenhouse Gas Initiative (RGGI), the longest-running U.S. carbon market, comes from comparisons of participating to non-participating states or from decompositions of observed emissions trends. These designs have taught us a great deal—most notably that much of the emissions decline in RGGI states is attributable to the shale-gas boom and the Great Recession rather than to the cap itself (Murray and Maniloff 2015)—but they struggle to isolate the policy’s own contribution from the many confounders that move with it.

Virginia offers a rare opportunity to make progress. It is the only U.S. state to have *entered*, *exited*, and is scheduled to *re-enter* a carbon cap-and-trade program within a five-year span. Virginia joined RGGI in January 2021 under the Virginia Clean Economy Act (Commonwealth of Virginia 2020); following a change in gubernatorial administration, the Air Pollution Control Board repealed the enabling regulation and the state left the program after the December 2023 auction (Virginia State Air Pollution Control Board 2023); and, after a further change in administration, Virginia is scheduled to re-enter in July 2026. Crucially, the proximate cause of the 2023 exit was political—a new governor with the authority to direct the rulemaking—rather than a response to electricity-market or macroeconomic conditions. This political origin makes the *timing* of the withdrawal plausibly exogenous to the electricity-price and emissions outcomes we study, which is precisely the condition that comparison-of-trends designs typically cannot satisfy.

This paper uses Virginia’s reversal to ask four questions. First, what was the causal effect of the December 2023 exit on retail electricity prices, wholesale prices, and carbon-dioxide emissions? Second, are the effects of *exiting* a carbon market symmetric to the effects of *entering* one, or does participation leave a durable imprint—through irreversible investment, for instance—that a withdrawal does not undo? Third, how much of any emissions change in Virginia leaks across state lines within the PJM Interconnection, the wholesale market in which Virginia sits? Fourth, does the scheduled 2026 re-entry reproduce the entry effects, providing an additional, prospective test?

Answering these questions requires a counterfactual Virginia. We construct one with the synthetic control method (Abadie et al. 2010; 2015; Abadie 2021), building a weighted combination of donor states—both non-RGGI states in PJM and states that remained in RGGI throughout—that tracks Virginia’s pre-treatment price and emissions trajectories. We complement the synthetic control with an event-study difference-in-differences specification and with synthetic difference-in-differences (Arkhangelsky et al. 2021), and we conduct inference appropriate to a single treated unit and a small number of clusters (Cattaneo et al. 2021; Cameron et al. 2008). The empirical content of the paper rests entirely on public data—retail prices from the U.S. Energy Information Administration, hourly emissions

from the EPA’s continuous emissions monitoring system, wholesale prices from PJM, and allowance prices from RGGI—so that the analysis is fully reproducible and does not depend on proprietary utility records.

The paper makes four contributions. It provides, to our knowledge, the first causal estimate of the effect of *removing* a carbon market, complementing a literature focused almost entirely on adoption. It exploits the entry/exit pairing to test for asymmetry and hysteresis in carbon-pricing effects—a question with direct bearing on whether temporary participation yields lasting benefits, and one that the policy-reversal literature has raised in other settings (Greenstone 2002; Walker 2013) but never tested for a carbon market. It delivers an event-based estimate of within-RTO leakage under sub-national pricing, sharpening cross-sectional estimates of RGGI leakage (Fell and Maniloff 2018). And it develops a paired entry/exit/re-entry research design on a single treated unit that may be of independent methodological interest for evaluating reversible policies.

The remainder of the paper is organized as follows. Section 2 describes RGGI and Virginia’s participation history. Section 3 situates the paper in the literature. Section 4 presents a simple model of how an allowance price enters retail and wholesale prices and derives the predictions we test. Section 5 describes the data and Section 6 the empirical strategy. Section 7 addresses threats to identification. Sections 8 and 9 report and interpret the results, and Section 10 concludes.

2 Institutional Background

2.1 RGGI Overview

RGGI is a cooperative cap-and-trade program covering carbon-dioxide emissions from fossil-fired electric generators in a group of Northeast and Mid-Atlantic states. Launched in 2009, it was the first mandatory market-based greenhouse-gas program in the United States. The program caps regional power-sector emissions and distributes nearly all allowances through quarterly auctions; covered generators must surrender one allowance per short ton of CO₂ emitted over each multi-year compliance period. Proceeds accrue to the participating states, which direct the bulk of revenue to energy-efficiency and clean-energy programs (Hibbard et al. 2023). Two price-stability mechanisms bound the allowance market: a Cost Containment Reserve (CCR) releases additional allowances when prices rise above a trigger, and an Emissions Containment Reserve (ECR) withholds allowances when prices fall below a floor; auctions also clear no lower than a minimum reserve price.

The program is periodically recalibrated through formal program reviews. The Third Program Review, finalized in 2025 and effective in 2027, sharply tightens the trajectory: the regional cap steps down by roughly ten and a half percent of the 2025 budget per year through 2033 and by about three percent per year thereafter, the minimum reserve price rises to \$9.00 in 2027 and escalates seven percent annually, the offset provisions are eliminated, and the ECR is replaced by an expanded two-tier CCR (RGGI, Inc. 2025b). These changes matter for our setting because they raise the expected stringency—and therefore the expected allowance price—of the program Virginia is scheduled to re-enter in 2026, a point we return to when

interpreting the re-entry design.

2.2 Virginia’s RGGI Timeline

Table 1: Virginia’s RGGI participation timeline.

Date	Event	Political context
Apr. 2020	Clean Economy Act & Community Flood Preparedness Act signed	Democratic trifecta
Jan. 2021	Virginia begins RGGI participation	First compliance period
Jan. 2022	Youngkin inaugurated; signals intent to withdraw	Republican governor, split legislature
Jun. 2023	Air Pollution Control Board votes to repeal RGGI regulation	Executive-driven withdrawal
Dec. 2023	Last Virginia participation in RGGI auction	Formal exit
Jan. 2024–Jun. 2026	Virginia outside RGGI	Non-participation period
2025	Spanberger (D) elected; legislation to rejoin signed	Democratic trifecta restored
Jul. 2026	Virginia re-enters RGGI	Second participation period

Sources: [Commonwealth of Virginia \(2020\)](#); [Virginia State Air Pollution Control Board \(2023\)](#); [RGGI, Inc. \(2025a\)](#).

2.3 The Dominion Energy Context

Virginia’s electricity market is dominated by Dominion Energy Virginia, a vertically integrated, rate-regulated utility. Unlike generators in restructured markets, Dominion recovers prudently incurred costs through retail rates set by the State Corporation Commission (SCC), and it recovered its RGGI allowance costs through a dedicated rate-adjustment clause, “Rider RGGI.” The SCC established the rider in docket PUR-2020-00169, approving a revenue requirement on the order of \$168 million for the first rate year, and later addressed its reinstatement and revision in subsequent dockets as allowance costs grew ([Virginia State Corporation Commission 2021](#)). Because the rider was a separately stated, cost-pass-through line item, Virginia’s exit removed an identifiable component of residential bills, and re-entry will restore one—an institutional feature we exploit to separate the mechanical (rider-driven) retail effect from any equilibrium effect operating through wholesale prices.

Three concurrent developments complicate the Virginia setting and motivate the controls and robustness checks below. Residential rates were rising over the sample period for reasons unrelated to RGGI; explosive data-center load growth in Northern Virginia was placing upward pressure on demand and, potentially, on prices ([Shehabi et al. 2024](#)); and the 2022 Inflation Reduction Act was changing the economics of clean generation nationwide. All three

move electricity outcomes and overlap in time with Virginia’s RGGI reversal, so a credible design must hold them constant. Throughout, we use only public data—SCC filings for the rider, and the public sources described in Section 5 for all outcomes; no proprietary Dominion records are used.

3 Literature Review

This paper draws on, and contributes to, four literatures.

The empirical effects of carbon pricing. A first strand estimates how carbon markets affect emissions and prices. For RGGI specifically, [Murray and Maniloff \(2015\)](#) decompose the post-2009 emissions decline and attribute the majority to falling natural-gas prices and the recession rather than to the cap, a finding that frames the identification challenge we confront. [Fell and Maniloff \(2018\)](#) document substantial emissions leakage to neighboring non-RGGI regions, and [Yan \(2021\)](#) reaches a related conclusion studying fossil-fuel consumption. On prices, recent work finds that RGGI adoption raised retail electricity prices in participating states by roughly ten percent in the short run ([Song and Hochman 2025](#)). Background on the earlier U.S. cap-and-trade experience for SO_2 and NO_x ([Burtraw and Szambelan 2009](#)) and on the theory of sub-national carbon pricing ([Shobe and Burtraw 2012](#)) rounds out this strand. What unites these studies is a focus on *adoption*; none estimates the effect of removing a carbon market.

Cost pass-through and incidence. A second strand asks how emissions costs are transmitted into electricity prices. In competitive wholesale markets, pass-through can be substantial and even exceed one in equilibrium ([Fabra and Reguant 2014](#)), and emission costs are reflected in hourly prices in restructured systems ([Hintermann 2016](#)). The incidence of regulation across firms can be large even when aggregate price effects are modest ([Bushnell et al. 2013](#)), and accounting for entry, exit, and investment dynamics changes the welfare assessment of market-based regulation ([Fowle et al. 2016](#)). This literature informs our model of how the allowance price enters Virginia’s regulated retail rate and PJM wholesale prices, and it cautions that the wholesale-price response need not equal the rider-driven retail response.

Policy reversal and asymmetry. A third, smaller strand studies what happens when environmental regulation is tightened or relaxed. [Greenstone \(2002\)](#) estimates the industrial-activity effects of Clean Air Act nonattainment designations, and [Walker \(2013\)](#) traces the labor-market adjustment costs of the resulting sectoral reallocation; [Coglianese et al. \(2020\)](#) disentangles the roles of fuel prices and regulation in the decline of U.S. coal. These papers establish that environmental regulation can have persistent, partly irreversible effects on the capital stock and workforce—the mechanism that would generate asymmetry between entry and exit—but the question has not been posed for a carbon market, where we can observe both directions of treatment on the same unit.

Methodology. Finally, the paper builds on the synthetic control method ([Abadie et al. 2010; 2015](#)) and its practical guidance ([Abadie 2021](#)), on synthetic difference-in-differences

(Arkhangelsky et al. 2021), and on modern difference-in-differences with multiple periods (Callaway and Sant’Anna 2021). For inference we use prediction intervals tailored to synthetic control (Cattaneo et al. 2021) and wild cluster bootstrap methods for settings with few clusters (Cameron et al. 2008). Our paired entry/exit design extends these tools to the evaluation of a reversible policy.

4 Theoretical Framework

To organize the empirical analysis we set out a deliberately simple partial-equilibrium account of how an allowance price enters the prices a Virginia consumer faces. The model is not meant to be estimated structurally; its purpose is to make explicit why the retail and wholesale responses to RGGI participation may differ, and why the response to exit need not mirror the response to entry.

4.1 A Carbon Adder in a Regulated Market

We model the retail price of electricity as

$$p_r = c_g + c_T + \pi_{\text{allow}} \cdot e + \mu, \quad (1)$$

where c_g is generation cost (fuel, O&M), c_T is transmission and distribution cost, π_{allow} is the allowance price (\$/MWh once scaled by the emissions rate e in tons/MWh of the marginal generator), and μ is the regulated markup. Removing the carbon adder ($\pi_{\text{allow}} \rightarrow 0$) reduces p_r mechanically through the rider; the wholesale-price effect depends on whether RGGI-state generators are marginal in PJM dispatch.

4.2 Predictions

The model yields five testable predictions, which map directly onto the outcomes and tests of Sections 6–9.

Prediction 1 (Retail prices). *Retail prices fall by approximately the rider amount on exit and rise by approximately the new rider on re-entry.*

Prediction 2 (Wholesale LMPs). *Ambiguous: Virginia generators offer lower prices, but if inframarginal the LMP may not move.*

Prediction 3 (Virginia emissions). *Rise if the adder was constraining dispatch of Virginia fossil generators.*

Prediction 4 (System-wide emissions). *Ambiguous: depends on whether Virginia generation displaces cleaner or dirtier out-of-state generation (leakage).*

Prediction 5 (Asymmetry/hysteresis). *Entry effects may exceed exit effects if 2021–2023 clean investments are irreversible, efficiency gains persist, or adjustments have ratchet properties.*

5 Data

We assemble a monthly panel from public sources only; Table 2 lists each series and its origin. Retail electricity prices by customer class come from EIA Form 861M (monthly) and Form 861 (annual). Wholesale prices are locational marginal prices (LMPs) from PJM’s Data Miner 2, aggregated from hourly to monthly. Carbon-dioxide emissions and gross load at the generating-unit level come from the EPA’s continuous emissions monitoring system, via the Clean Air Markets Program Data portal, aggregated to the plant, state, and PJM-zone levels. Allowance prices and quantities come from RGGI’s auction records (RGGI, Inc. 2025a). We control for the principal drivers of electricity prices and emissions: natural-gas prices (EIA Henry Hub and the regionally relevant Transco Zone 5 hub), heating- and cooling-degree days (NOAA), and state economic conditions (income and population from BEA and the Census). The Rider RGGI cost passed to Virginia customers is taken from public SCC filings (Virginia State Corporation Commission 2021).

Table 2: Data sources (public). **[TODO: Regenerated from config in Phase 2–3.]**

Variable	Source	Frequency	Coverage
Retail electricity prices	EIA Form 861/861M	Monthly/Annual	States, 2018–2026
Wholesale prices (LMPs)	PJM Data Miner 2.0	Hourly → monthly	PJM nodes/zones
CO ₂ emissions	EPA CEMS (CAMPD)	Hourly → monthly	U.S. fossil units
Generation by fuel	EIA 923 / EPA CEMS	Monthly	U.S. generators
Allowance auction prices	RGGI Inc. reports	Quarterly	2008–present
Natural gas prices	EIA (Henry Hub, Transco Z5)	Daily/Monthly	National/Regional
Weather (HDD/CDD)	NOAA ISD/GHCN	Daily	By state
Economic controls	BEA / Census	Annual/Quarterly	States
Dominion rider	Public SCC dockets	Per filing	Virginia

Notes: Public sources only; no Dominion internal data. **[TODO: Add access dates/vintages in Phase 2.]**

5.1 Sample Construction

The unit of analysis is the state-month for the price and emissions panels, and the PJM-zone month for the wholesale analysis; the sample window is January 2018 through mid-2026, spanning the pre-entry baseline, the participation period, and the post-exit period. Virginia is the treated unit. We use two donor pools. The first comprises non-RGGI states in the PJM Interconnection—West Virginia, Ohio, Pennsylvania, Kentucky, Indiana, North Carolina, and Tennessee—which share Virginia’s wholesale market but never priced carbon. The second

comprises states that participated in RGGI continuously throughout the sample—Maryland, Delaware, New Jersey, Connecticut, Massachusetts, Maine, New Hampshire, Vermont, Rhode Island, and New York. The two pools serve complementary purposes: the non-RGGI PJM states isolate the effect of carbon pricing within a common wholesale market, while the continuous-RGGI states absorb shocks common to carbon-pricing jurisdictions. Synthetic control selects, data-driven, the weighted combination of donors that best reproduces Virginia’s pre-treatment path.

6 Empirical Strategy

6.1 Synthetic Control (Primary)

Our primary estimator is the synthetic control method. For each event and outcome we construct a synthetic Virginia as a convex combination of donor states whose weights are chosen to minimize the discrepancy between Virginia and the synthetic unit over the pre-treatment period, matching on pre-treatment outcome values and on the covariates listed in Section 5. The treatment effect in each post-treatment month is the gap between observed Virginia and its synthetic counterpart. We estimate the method separately for the two realized events: entry, with a pre-treatment window of January 2018 through December 2020 and a post-treatment window running to the exit; and exit, with a pre-treatment window of January 2021 through the months preceding the December 2023 withdrawal and a post-treatment window running through the end of the sample. Inference proceeds by permutation: we reassign the treatment to each donor in turn, compute the resulting placebo gaps, and assess whether Virginia’s gap is extreme relative to the placebo distribution (Abadie et al. 2010; Abadie 2021). We supplement the permutation test with the conformal prediction intervals of Cattaneo et al. (2021).

6.2 Event-Study Difference-in-Differences

$$Y_{st} = \alpha_s + \gamma_t + \sum_{\tau \neq -1} \beta_\tau \mathbf{1}[s = \text{VA}] \mathbf{1}[t - t^* = \tau] + X'_{st} \delta + \varepsilon_{st}, \quad (2)$$

with state fixed effects α_s , month fixed effects γ_t , time-varying controls X_{st} (gas prices, HDD, CDD), and dynamic effects β_τ relative to $\tau = -1$. Standard errors clustered at the state level; wild cluster bootstrap given the small number of clusters. Estimated separately for the exit ($t^* = \text{Dec. 2023/Jan. 2024}$), entry ($t^* = \text{Jan. 2021}$), and prospectively re-entry ($t^* = \text{Jul. 2026}$) events.

6.3 Synthetic Difference-in-Differences

As a bridge between the two approaches we report synthetic difference-in-differences (Arkhangelsky et al. 2021), which combines the unit reweighting of synthetic control with the double-differencing of difference-in-differences and additionally reweights time periods. It is more robust than synthetic control when pre-treatment fit is imperfect and more robust than two-way fixed effects when the treated unit is not well represented by a simple average of

controls, so agreement across the three estimators provides reassurance that the results are not an artifact of any single method.

6.4 Extensions

We pursue four extensions. First, we replace the state panel with a PJM-zone panel to study wholesale prices, treating the Dominion (DOM) zone as the treated unit and other PJM zones as controls. Second, we estimate emissions effects both for Virginia generators and for PJM as a whole; the difference identifies the leakage rate—the share of any Virginia emissions change offset by changes elsewhere in the interconnection. Third, we formally test for asymmetry by comparing the paired estimates, $|\hat{\tau}^{\text{exit}}| = |\hat{\tau}^{\text{entry}}|$; a rejection is evidence of hysteresis. Fourth, where public microdata permit, we examine heterogeneity in the retail-price effect across customer classes.

7 Threats to Identification and Robustness

The design’s credibility rests on the assumption that, absent the policy change, Virginia’s outcomes would have evolved like those of its synthetic counterpart. Several threats could violate this assumption, and we address each with a corresponding check. Table 3 summarizes them.

The most important threat is *anticipation*. Governor Youngkin signaled his intent to withdraw from RGGI shortly after taking office in January 2022, well before the formal exit in December 2023, so forward-looking generators and investors may have adjusted in advance. We address this by re-estimating with alternative treatment dates—the January 2022 announcement, the June 2023 Board vote, and the December 2023 final auction—and by inspecting the pre-treatment gaps for evidence of anticipatory breaks. A second concern is *concurrent policy shocks*: the Inflation Reduction Act and changes in state renewable-portfolio standards moved clean-energy investment over the same period. We control for renewable capacity additions and federal-subsidy proxies and show that the estimates are stable to their inclusion. Third, *gas-price volatility* drives both prices and the dispatch order; we control for regional gas-hub prices and report gas-adjusted outcomes. Fourth, *data-center load growth* in Northern Virginia could confound price effects; we control for large-customer load and use per-megawatt-hour prices rather than total revenue. Fifth, the within-PJM design raises a *SUTVA* concern, since Virginia’s exit may affect other PJM states; we estimate PJM-wide wholesale effects and bound the magnitude of spillovers. Finally, because the rider is a mechanical pass-through, we decompose the retail effect into its rider and non-rider (equilibrium) components, and we control for fleet composition to ensure that capacity additions and retirements rather than the policy are not driving the results. Inference throughout uses permutation tests for the synthetic control and the wild cluster bootstrap for the difference-in-differences estimates, both appropriate to the small number of treated units and clusters (Cattaneo et al. 2021; Cameron et al. 2008).

Table 3: Threats and robustness checks.

Threat	Test / mitigation
Anticipation (exit signaled early 2022)	Alternative treatment dates (Jan. 2022, Jun. 2023, Dec. 2023); pre-trend tests.
Concurrent policy shocks (IRA, RPS)	Control for renewable capacity additions and federal subsidy proxies.
Gas price volatility	Control for regional gas hub prices; gas-adjusted outcomes.
Data center load growth	Control for large-customer load; use per-MWh prices.
SUTVA violation (PJM spillovers)	Estimate PJM-wide wholesale effects; bound spillovers.
Small- N inference	Permutation inference (SCM); wild cluster bootstrap (DiD).
Rider mechanicality	Decompose retail effect into rider (mechanical) and non-rider components.
Fleet composition changes	Control for capacity additions/retirements; shares vs. levels.

8 Results

[PENDING DATA: All results below are generated by scripts/R/ from public data and pulled in via \input/\includegraphics. No numbers are hand-entered.]

8.1 Exit Event (Dec. 2023)

[PENDING DATA: SCM gap plot, event-study coefficients, placebo distribution.]

8.2 Entry Event (Jan. 2021)

[PENDING DATA: SCM and event-study estimates for entry.]

8.3 Asymmetry Test

[PENDING DATA: Formal test of $|\hat{\tau}^{\text{exit}}| = |\hat{\tau}^{\text{entry}}|$.]

8.4 Emissions Leakage

[PENDING DATA: VA-plant vs. PJM-wide emissions; implied leakage rate.]

9 Expected Results and Interpretation

[TODO: Filled alongside results.] We pre-register three scenarios:

Scenario A — Symmetric effects. Carbon-pricing effects are reversible; policy uncertainty may undermine long-run investment incentives.

Scenario B — Asymmetric (entry > exit). Irreversible clean investments create a ratchet; even temporary participation yields lasting benefits.

Scenario C — Leakage dominates. Sub-national design reshuffles generation across state lines without net reductions.

Re-entry (prospective). [PENDING DATA: The Jul. 2026 re-entry is in the future as of this draft. We pre-register the design (treatment date, donor pool, estimands, inference) and will report estimates once post-treatment data are available.]

10 Conclusion

[TODO: Written after results.]

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A Data Sources and Public Retrieval Methods

[TODO: Phase 3 – map each variable to its public source and access method.]

B Pre-Analysis Plan: Re-Entry (Jul. 2026)

[TODO: Phase 4 – pre-registered estimands, treatment date, donor pool, and inference for the re-entry event, fixed in advance of observing outcomes.]